

What is Normalization ?

" It is the process of reducing a large table into smaller tables in order to remove redundancies and anomalies by identifying their functional dependencies is known as Normalization . "

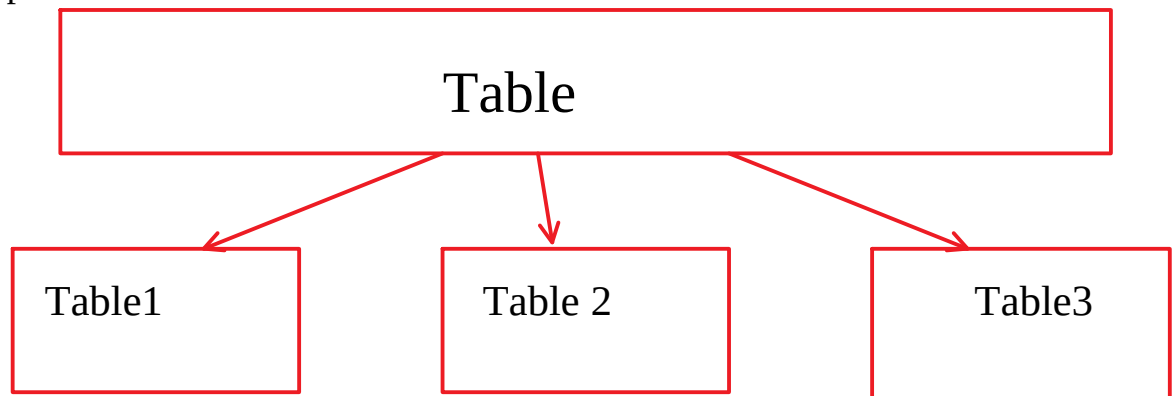
Or

"The process of decomposing a large table into smaller table is known as Normalization."

Or

Reducing a table to its **Normal Form** is known as Normalization . ""

Example



What is Normal Form ?

A table without redundancies and anomalies are said to be in Normal Form .

Levels of Normal Form .

1. **First Normal Form (1NF)**
2. **Second Normal Form (2NF)**
3. **Third Normal Form (3NF)**
4. **Boyce - Codd Normal Form (BCNF 3.5NF)**
5. **Fifth normal form (5F)**
6. **Sixth normal form (6F)**

Note : If any Table / entity is reduced to 3NF , then the table is said to be normalized.

1. **First Normal Form (1NF) :**

The table is said to be first normal form if it is satisfied following conditions

- No duplicates/repeated records .
- Multivalued data should not be present .
- each and every cell of table should store a single value
- column names must be unique
- rows of the table must be unique

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<u>QID</u>	<u>NAME</u>	<u>COURSE</u>
1	A	JAVA
2	B	JAVA , SQL
3	C	MT , SQL
1	A	MT

<u>QID</u>	<u>NAME</u>	<u>C1</u>	<u>C2</u>	<u>C3</u>
1	A	JAVA		MT
2	B	JAVA	SQL	
3	C		SQL	MT

1. **Second Normal Form (2NF)**

The table is said to be second normal form if it is satisfied following conditions

- Table should be in 1NF
- Table should not have Partial Functional Dependency .

EMPLOYEE - (EID , ENAME , SAL , DEPTNO , DNAME ,LOC)

EID	ENAME	SAL	DEPTNO	DNAME	LOC
1	A	100	10	D1	L1
2	B	120	20	D2	L2

3	C	320	10	D1	L1
4	D	251	10	D1	L1

EID- ename,
sal DEPTNO
– dname, loc

(Eid, deptno) **7** (ename, sal, dname, loc) composite key
attribute

Result

s in PFD R1 (eid, ename,sal)

R1 – (eid, ename,sal)

R2 – (deptno, dname, loc)

<u>Eid</u>	<u>ename</u>	<u>sal</u>	<u>DNO</u>
1	A	100	10
2	B	120	20
3	C	320	10
4	D	251	10

<u>Deptno</u>	<u>dname</u>	<u>Loc</u>
10	D1	L1
20	D2	L2

1. **Third Normal Form (3NF)**

The table is said to be third normal form if it is satisfied following conditions

- Table should be in 2NF .
- Table should not have Transitive Functional Dependency .

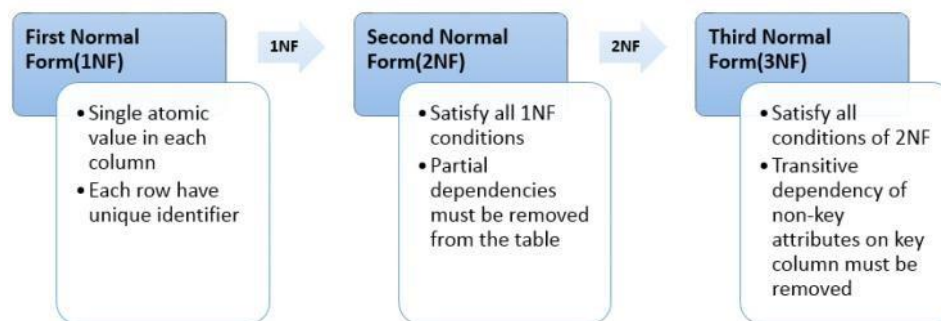
Employee - (**EID** , Ename , Sal , comm , Pincode , state , country)

EID -> ENAME
SAL

COMM
PINCODE -> STATE
COUNTRY

R1 – (eid,
ename, comm) R2 –
(pincode, state,
country)

Note



What is a trigger?(PL/SQL)

A DB trigger is a code or programs that automatically execute with response to some event on a table or view in a database.

Example: When a new student is added to the student database, new records should be created in the related tables like Exam, Score and Attendance tables.

A Trigger is a SQL procedure that initiates an action in response to an event (Insert, Delete or Update) occurs. When a new Employee is added to an Employee_Details table, new records will be created in the relevant tables such as Employee_Payroll, Employee_Time_Sheet etc.,

Syntax:-

```
create trigger [trigger_name]
[before | after]
{insert | update | delete}
on [table_name]
[for each row]
[trigger_body]
```

create trigger [trigger_name]: Creates or replaces an existing trigger with the trigger_name.

[before | after]: This specifies when the trigger will be executed.

{insert | update | delete}: This specifies the DML operation.

on [table_name]: This specifies the name of the table associated with the trigger.

[for each row]: This specifies a row-level trigger, i.e., the trigger will be executed for each row being affected.

[trigger_body]: This provides the operation to be performed as trigger is fired

Example:-

```
create trigger stud_marks
before INSERT
on
Student
for each row
set Student.total = Student.subj1 + Student.subj2 + Student.subj3, Student.per = Student.total * 60 / 100;
```

What is an Index?(PL/SQL)

An index is performance tuning method of allowing faster retrieval of records from the table. An index creates an entry for each value and it will be faster to retrieve data.

Syntax:-

Create index index_name on table_name(column_name);

Example:- create index i1 on gmail(username);

Select * from gmail where username='abc@gmail.com'; //1 out 1